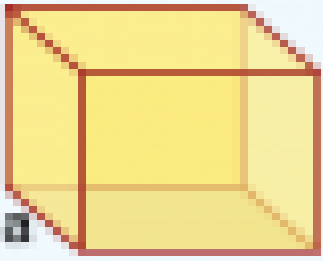
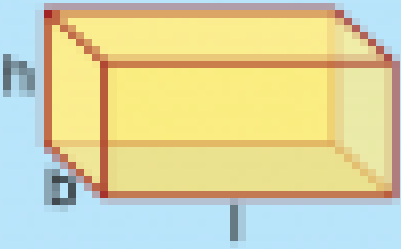
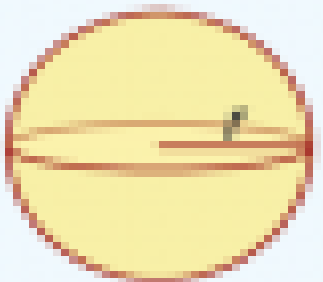




Mensuration – 3D

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Solid	Volume	CSA	TSA	Figure
Cube a = side	a^3	$4 a^2$	$6 a^2$	
Cuboid l = length b = breadth h = height	$l \times b \times h$	$2h(l + b)$	$2 (lb +bh +hl)$	
Sphere r = radius	$(4/3) \pi r^3$	$4 \pi r^2$	$4 \pi r^2$	
Hemisphere r = radius	$(3/2) \pi r^3$	$2 \pi r^2$	$3 \pi r^2$	
Cylinder r = radius h = height	$\pi r^2 h$	$2\pi r h$	$2\pi rh + 2\pi r^2$	
Cone r = radius l = slant height h = height	$(3/2) \pi r^2 h$	$\pi r l$	$\pi r (r + l)$	

If two cubes of sides 9 cm each are joined end by end then the surface area of cuboid will

A)

☐ 800 cm^2

B)

☐ 810 cm^2

C)

☐ 820 cm^2

D)

☐ 830 cm^2

If the area of base of a cone is 154 cm^2 and height is 9 cm , then the curved surface area of cone will be

A)

☐ 200.53 cm^2

B)

☐ 220.53 cm^2

C)

☐ 240.8 cm^2

D)

☐ 250.8 cm^2

The number of cubical ice pieces of side 3 cm that can be cut from a cube of ice of side 9 cm is

A)

☐ 16

B)

☐ 26

C)

☐ 27

D)

☐ 28

The bricks of dimension $15\text{cm} \times 25\text{cm} \times 7\text{cm}$ required to build a wall $(3.5 \times 10 \times 12) \text{ m}^3$ are

A)

☐ 160000

B)

☐ 170000

C)

☐ 180000

D)

☐ 200000

How many cubes each of edge 6 cm can be cut from a cuboid of $42\text{cm} \times 36\text{cm} \times 24\text{cm}$?

A)

☐ 142

B)

☐ 186

C)

☐ 168

D)

☐ 124

The surface area of a cube is 216 sq. cm. Find its volume

A)

☐ **216 cm^3**

B)

☐ **150 cm^3**

C)

☐ **136 cm^3**

D)

☐ **263 cm^3**

Volume of two cubes are in the ratio 1 : 27, find the ratio of their surface areas.

A)

☐ 2 : 9

B)

☐ 1 : 9

C)

☐ 1 : 11

D)

☐ 2 : 13

A rectangular piece of paper is 71 cm long and 10 cm wide. A cylinder is formed by rolling the paper along its breadth. Find the volume of the cylinder.

A)

☐ 654

B)

☐ 764

C)

☐ 500

D)

☐ 564.78

Two circular cylinders of equal volume have their heights in the ratio 9 : 16 Find the ratio of their radii

A)

☐ 4 : 3

B)

☐ 5 : 6

C)

☐ 2 : 5

D)

☐ 5 : 7

How many coins 1.4 cm in diameter and 0.4 cm thick is to be melted to form, right circular cylinder of height 16 cm and diameter 3.5 cm

A)

☐ 200

B)

☐ 150

C)

☐ 250

D)

☐ 300

A conical tent is of diameter 12m at the base and its height is 8m. Find the area of the canvas (in m^2) required to make the tent.

A)

☐ 188.57 m

B)

☐ 288 m

C)

☐ 400 m

D)

☐ 500 m

Two right circular cones of equal curved surface areas have their slant heights in the ratio of 3 : 5. Find the ratio of their radii.

A)

☐ 5 : 3

B)

☐ 5 : 7

C)

☐ 8 : 3

D)

☐ 3 : 5

Find the number of lead balls of radius 1 cm each that can be made from a sphere of radius 4 cm.

A)

☐ 24

B)

☐ 85

C)

☐ 64

D)

☐ 46

The curved surface area of a right circular cylinder of height 14 cm is 88 cm^2 . Find the diameter of the base of the cylinder.

A)

☐ 2 cm

B)

☐ 5cm

C)

☐ 4 cm

D)

☐ 6 cm

The ratio between the length and width of a field is 10 : 60 The area of the 3840 m^2 . Find the difference between the length and width of the field.

A)

☐ 32 m

B)

☐ 24 m

C)

☐ 20 m

D)

☐ 18 m